

APPENDIX 1: DEFINITIONS OF TASK MEASURES FROM THE 1977 DICTIONARY OF OCCUPATIONAL TITLES

Variable	DOT definition	Task interpretation	Example tasks from <i>Handbook for Analyzing Jobs</i>
1. GED Math (MATH)	General educational development, mathematics	Measure of nonroutine analytic tasks	Lowest level: Adds and subtracts 2-digit numbers; performs operations with units such as cup, pint, and quart. Midlevel: Computes discount, interest, profit, and loss; inspects flat glass and compiles defect data based on samples to determine variances from acceptable quality limits. Highest level: Conducts and oversees analyses of aerodynamic and thermodynamic systems . . . to determine suitability of design for aircraft and missiles.
2. Direction, Control, Planning (DCP)	Adaptability to accepting responsibility for the direction, control, or planning of an activity	Measure of nonroutine interactive tasks	Plans and designs private residences, office buildings, factories, and other structures; applies principles of accounting to install and maintain operation of general accounting system; conducts prosecution in court proceedings . . . gathers and analyzes evidence, reviews pertinent decisions . . . appears against accused in court of law; commands fishing vessel crew engaged in catching fish and other marine life.
3. Set Limits, Tolerances, or Standards (STS)	Adaptability to situations requiring the precise attainment of set limits, tolerances, or standards	Measure of routine cognitive tasks	Operates a billing machine to transcribe from office records data; calculates degrees, minutes, and second of latitude and longitude, using standard navigation aids; measures dimensions of bottle, using gauges and micrometers to verify that setup of bottle-making conforms to manufacturing specifications; prepares and verifies voter lists from official registration records.
4. Finger Dexterity (FINGDEX)	Ability to move fingers, and manipulate small objects with fingers, rapidly or accurately	Measure of routine manual tasks	Mixes and bakes ingredients according to recipes; sews fasteners and decorative trimmings to articles; feeds tungsten filament wire coils into machine that mounts them to stems in electric light bulbs; operates tabulating machine that processes data from tabulating cards into printed records; packs agricultural produce such as bulbs, fruits, nuts, eggs, and vegetables for storage or shipment; attaches hands to faces of watches.
5. Eye Hand Foot Coordination (EYEHAND)	Ability to move the hand and foot coordinately with each other in accordance with visual stimuli	Measure of nonroutine manual tasks	Lowest level: Tends machine that crimps eyelets, grommets; next level: attends to beef cattle on stock ranch; drives bus to transport passengers; next level: pilots airplane to transport passengers; prunes and treats ornamental and shade trees; highest level: performs gymnastic feats of skill and balance.

Source: U. S. Department of Labor, Manpower Administration, *Handbook for Analyzing Jobs* (Washington, DC, 1972).

APPENDIX 2: COMPUTERIZATION AND INDUSTRY TASK INPUT, 1960–1998:
 USING COMPOSITE TASK MEASURES
 DEPENDENT VARIABLE: $10 \times$ ANNUAL WITHIN-INDUSTRY CHANGE IN TASK INPUT,
 MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

		1. 1990– 1998	2. 1980– 1990	3. 1970– 1980	4. 1960– 1970
A. Δ Nonroutine analytic	Δ Computer use	8.21	12.09	6.50	8.57
	1984–1997	(4.55)	(4.55)	(4.21)	(5.74)
	Intercept	0.64	–0.67	0.07	–0.07
		(0.96)	(0.94)	(0.86)	(1.14)
	R^2	0.02	0.05	0.02	0.02
B. Δ Nonroutine interactive	Weighted mean Δ	2.26	1.67	1.32	1.51
	Δ Computer use	9.83	9.93	5.67	10.45
	1984–1997	(4.39)	(4.74)	(3.47)	(5.00)
	Intercept	0.54	0.53	1.42	0.00
		(0.92)	(0.98)	(0.71)	(1.00)
C. Δ Routine cognitive	R^2	0.04	0.03	0.02	0.03
	Weighted mean Δ	2.48	2.45	2.51	1.93
	Δ Computer use	–13.40	–4.59	–4.76	–7.02
	1984–1997	(4.44)	(5.58)	(3.70)	(4.75)
	Intercept	0.23	–0.39	0.56	–0.05
D. Δ Routine manual		(0.94)	(1.16)	(0.76)	(0.95)
	R^2	0.06	0.00	0.01	0.02
	Weighted mean Δ	–2.41	–1.28	–0.35	–1.35
	Δ Computer use	–23.17	–15.27	–13.25	3.98
	1984–1997	(6.99)	(6.86)	(5.34)	(4.13)
	Intercept	–0.52	–0.80	1.97	1.35
		(1.47)	(1.42)	(1.09)	(0.82)
	R^2	0.07	0.03	0.04	0.01
	Weighted mean Δ	–5.09	–3.76	–0.56	2.09

n is 140 consistent CIC industries. Standard errors are in parentheses. Each column of panels A–D presents a separate OLS regression of ten times the annual change in industry-level task input between the endpoints of the indicated time interval (measured in centiles of the 1960 task distribution) on the annual percentage point change in industry computer use during 1984–1997 (mean 0.193) and a constant. Computer use is the fraction of industry workers using a computer at their jobs, estimated from the October 1984 and 1997 CPS samples. Estimates are weighted by mean industry share of total employment in FTEs over the endpoints of the years used to form the dependent variable. Samples used are Census 1960, 1970, and 1980 and CPS MORG 1980, 1990, and 1998. See the Data Appendix for details on construction of the composite task variables. See Table I for definitions and examples of task variables.

DATA APPENDIX

A.1. Samples Used from Current Population Survey and Census of Populations

To calculate occupational and education distributions economywide and within industries for 1960 to 1998, we used observations on all noninstitutionalized, employed workers ages

18–64 from the Census PUMS one percent samples for 1960, 1970, 1980, and 1990 [Ruggles and Sobek, 1997] and the Merged Outgoing Rotation Groups of the Current Population Survey for the years 1980, 1990, and 1998. All individual and industry level analyses are performed using as weights full-time equivalent hours of labor supply, which is the product of the individual Census or CPS sampling weight times hours of work in the sample reference week divided by 35 and, for Census samples, weeks of work in the previous year. Because hours are not reported for the self-employed in the CPS prior to 1994, we assigned self-employed workers in all CPS samples the average labor hours in their industry-education-year cell. In cases where industry hours supplied by education category were unavailable (due to an empty industry-education-year cell), we assigned weekly hours as the mean of workers' education-year cells.

To attain comparable educational categories across the redefinition of Census Bureau's education variable introduced in 1990 in the Census and in 1992 in the CPS, we use the method proposed by Jaeger [1997]. In data coded with the pre-1992 education question (Census PUMS 1960, 1970, and 1980, and CPS MORG files 1980 and 1990), we defined high school dropouts as those with fewer than twelve years of completed schooling; high school graduates as those having twelve years of completed schooling; some college attendees as those with any schooling beyond twelve years (completed or not) and fewer than sixteen completed years; and college plus graduates as those with sixteen or more years of completed schooling. In data coded with the revised education question (1990 Census PUMS and 1998 CPS MORG file), we define high school dropouts as those with fewer than twelve years of completed schooling, high school graduates as those with either twelve completed years of schooling or a high school diploma or G.E.D.; some college as those with some college or holding an Associate's Degree; and college plus as those with a B.A. or higher.

A.2. Computing DOT Task Means for Census Occupation Categories (COCs)

To compute DOT Task Means for 1970 CIC Occupations, we used the April 1971 CPS Monthly File issued by the National Academy of Sciences [1981] in which experts assigned individual DOT occupation codes and associated DOT measures to each of 60,441 workers. Because Census occupation categories are sig-

nificantly coarser than DOT occupation categories, the 411 1970 census occupation codes represented in the 1971 CPS were assigned a total of 3886 unique 1977 DOT occupations. We used the CPS sampling weights to calculate means of each DOT task measure by occupation. Because the gender distribution of DOT occupations differs substantially within COC occupation cells, we performed this exercise separately by gender. In cases where a COC cell contained exclusively males or females, we assigned the cell mean to both genders. This provided a set of 822 DOT occupation means by 1970 COC and gender.

To generate DOT means for 1960 occupations, we developed a crosswalk from the 1970 to 1960 COC occupational classification schemes using information in Priebe and Greene [1972]. Our crosswalk (available on request) provides a set of 211 consistent 1960–1970 occupations representing the lowest common level of aggregation needed to obtain a consistent series. We applied the 1970 COC means to our 1970 Census sample by occupation and gender and calculated weighted gender-occupation means across the 211 consistent 1960–1970 occupational categories.

It was not possible to develop a bridging crosswalk between 1970 and 1980 COC occupations due to the substantial differences between these classifications. Instead, we employed a Census sample prepared for the Committee on Occupational Classification and Analysis chaired by Donald Treiman and provided to us by Michael Handel. This file contains 122,141 observations from the 1980 Census that are individually dual coded with both 1970 and 1980 COC occupation codes based on occupational and other demographic information supplied by Census respondents. To calculate DOT means by 1980 occupation, we merged the 1970 COC-DOT means (above) to the Treiman file by gender and 1970 COC occupation, achieving a 97 percent match rate. We appended to the Treiman file consistent occupation codes for the years 1980 to 1998 developed by Autor, Katz, and Krueger [1998], and calculated weighted means of each DOT measure within occupation-gender categories. This yielded DOT means by gender for each of 485 DOT occupations.

A.3. Computing DOT Task Means by Consistent 1960–1998 Industry

To compute DOT task means overall, by industry, and by industry-education cell for 1960–1998, we assigned the consistent DOT occupational task means for 1960–1998 by gender and